Quant Guide
TGIR QuantLib Tools

TG Investments and Research

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Contents

1 Scope

This guide is for quantitative users who need to understand the pricing assumptions, calibration logic, risk measures, and research references used in the workstation.

2 Portfolio Scope

The repo prices five instruments:

- a spot-starting fixed-versus-SOFR swap,
- a European swaption under a normal-volatility convention priced with Hull-White 1F plus Jamshidian's decomposition,
- a Bermudan swaption under a calibrated short-rate model,
- a second Bermudan swaption matching the workbook benchmark trade, and
- an SPX cliquet under a Black-Scholes-Merton process with analytic cliquet valuation.

3 Rates Curve Construction

3.1 Inputs

The rates market state follows the workbook strip and a configurable valuation date, defaulting to `2026-03-10`:

`1D`	`1W`	`2W`	`3W`	`1M`	`2M`	`3M`	`6M`	`9M`	`1Y`	`...30Y`

3.2 Bootstrap

The curve is constructed in `portfolio.py` with:

- one `ql.DepositRateHelper` for the `1D` point, and
- a strip of `ql.OISRateHelper` objects for the remaining term SOFR OIS points.

The term structure is a `ql.PiecewiseLogCubicDiscount`. Calibration quality is checked by rebuilding quoted OIS swaps and asserting near-zero NPV.

4 European Swaption Model

The European swaption calibrates `ql.HullWhite` to the selected ATM normal-volatility matrix pillar and prices with `ql.JamshidianSwaptionEngine`. The matrix follows the workbook surface, with short expiries from `1M` onward, annual expiries through `10Y`, then `12Y`, `15Y`, `20Y`, and `25Y`, against swap tenors through `30Y`. The on-screen matrix uses the exact workbook axes. When the Bermudan calibration needs a `3Y` expiry, it is linearly interpolated between the workbook `2Y` and `4Y` rows.

5 Bermudan Swaption Model

5.1 Calibration

The Bermudan stack:

1. identifies the trade's Bermudan call schedule and the remaining swap tenor implied by the common fixed final maturity,
2. creates one `ql.SwaptionHelper` per diagonal pillar,
3. calibrates either `ql.HullWhite` or `ql.G2` while holding the mean-reversion input fixed, and
4. prices the Bermudan with `ql.TreeSwaptionEngine`.

All rates trades are modeled as `ql.OvernightIndexedSwap` instruments with daily compounded SOFR coupons over the chosen reset period. Payment frequency and reset frequency are explicit trade inputs; the workbook reference trade uses annual pay and annual reset. For example, a `5Y NC 2Y` Bermudan keeps the same final maturity date and therefore exercises into the remaining `3Y`, `2Y`, and `1Y` swaps along its annual call schedule. The workstation defaults Bermudans to Hull-White 1F, and the detail page allows a switch to G2++ so users can compare values and risk outputs under the same market data.

5.2 Interpretation

The calibration helpers are labeled by the booked Bermudan schedule, for example `1Y x 4Y`, `2Y x 3Y`, `3Y x 2Y`, `4Y x 1Y` for a five-year annual Bermudan. When a booked expiry is not on the editable matrix axes, the helper linearly interpolates in the expiry dimension and keeps the exact source matrix points for audit purposes.

6 SPX Cliquet Model

6.1 State Variables

The cliquet reads:

- SPX spot,
- flat dividend yield,
- flat Black volatility,
- quantity,
- percentage moneyness,
- maturity in months, and
- reset frequency in months.

6.2 Process And Engine

The market process is a `ql.BlackScholesMertonProcess` with:

- the shared SOFR curve used as the risk-free curve,
- a flat dividend curve, and
- a flat Black volatility surface.

The instrument is a `ql.CliquetOption` with `ql.PercentageStrikePayoff` and `ql.AnalyticCliquetEngine`.

6.3 Extra Analytics

The cliquet page supplements the QuantLib price with:

- reset-period decomposition into forward-start option values,
- deterministic spot-volatility scenarios,
- trade-level analytic Greeks, and
- Monte Carlo payoff-distribution diagnostics.

7 Risk Outputs

7.1 Rates Trades

The swap and swaptions expose on-demand point sensitivities:

- curve ladder from single-point SOFR quote bumps,
- European swaption vega from a single matrix-cell bump,
- Bermudan matrix sensitivity from the trade schedule's helper source points, and
- a separate short-rate mean-reversion sensitivity for European and Bermudan swaptions.

7.2 Cliquet Trade

The cliquet page reports:

- delta, gamma, vega, theta, rho, and dividend rho,
- scenario NPVs for spot and volatility shocks, and
- distribution metrics such as 5th percentile PV and expected shortfall.

8 Validation

The test suite validates the pricing stack in three distinct ways:

- calibration repricing for the SOFR OIS curve,
- calibration repricing for Bermudan helpers implied by the fixed-maturity call schedule, and
- ten cliquet identity cases where the payoff reduces to simpler known structures.

9 Research References

The curated references live in `docs/RESEARCH.md` and on the in-app model page. The swaption list emphasizes:

- Peter Caspers' QuantLib-linked swaption implementation notes,
- Jamshidian's bond-option decomposition, and
- classic Bermudan exercise papers.

The cliquet list covers numerical methods, hedging, Levy models, regime-switching, and counterparty-risk extensions.

10 Model Boundaries

This workstation deliberately avoids:

- live data feeds,
- smile or skew modeling for the cliquet,
- short-rate models beyond the implemented Hull-White 1F and G2++ calibration paths,
- database-backed trade lifecycle, and
- portfolio aggregation beyond the five-trade sandbox.